MATH 211 Analysis III — Problem Session 5

Notation. For a function $f(x,y)$, the *partial derivative* of f with respect to x is defined by

$$f_x(x,y) = \frac{\partial f}{\partial x}(x,y) = \lim_{h\to 0} \frac{f(x+h,y) - f(x,y)}{h},$$

and with respect to y by

$$f_y(x,y) = \frac{\partial f}{\partial y}(x,y) = \lim_{k\to 0} \frac{f(x,y+k) - f(x,y)}{k}.$$

The two symbols f_x and $\dfrac{\partial f}{\partial x}$ denote exactly the same object (likewise f_y and $\dfrac{\partial f}{\partial y}$), and we use them interchangeably throughout. In practice, computing f_x means differentiating f with respect to x while treating y as a constant.

Problem 1

$f(x,y) = x\,e^y$. Compute $f_{xx}, f_{yy}, f_{xy}, f_{yx}$ and verify $f_{xy} = f_{yx}$.

Solution.

$$f_x = e^y, \qquad f_y = x\,e^y, \qquad f_{xx} = 0, \qquad f_{yy} = x\,e^y.$$

$$f_{xy} = \frac{\partial}{\partial y}\big(e^y\big) = e^y, \qquad f_{yx} = \frac{\partial}{\partial x}\big(x\,e^y\big) = e^y.$$

Problem 2

$f(x,y) = \cos(xy)$. Compute the mixed partials and verify $\dfrac{\partial^2 f}{\partial y\,\partial x} = \dfrac{\partial^2 f}{\partial x\,\partial y}$.

Solution.

$$\frac{\partial f}{\partial x} = -y\sin(xy), \qquad \frac{\partial f}{\partial y} = -x\sin(xy).$$

$$\frac{\partial^2 f}{\partial y\,\partial x} = -\sin(xy) - xy\cos(xy), \qquad \frac{\partial^2 f}{\partial x\,\partial y} = -\sin(xy) - xy\cos(xy).$$

Problem 3

$f(x,y) = \dfrac{x+y}{x^2+1}$. Compute the mixed partials and verify $f_{xy} = f_{yx}$.

Solution.

$$f_x = \frac{(x^2+1) - (x+y)(2x)}{(x^2+1)^2} = \frac{1 - x^2 - 2xy}{(x^2+1)^2}, \qquad f_y = \frac{1}{x^2+1}.$$

$$f_{xy} = \frac{-2x}{(x^2+1)^2}, \qquad f_{yx} = -(x^2+1)^{-2}\cdot 2x = \frac{-2x}{(x^2+1)^2}.$$

Problem 4

$f(x,y) = x^2 + \sin(xy)$. Compute f_x, f_y and state where they are continuous.

Solution.

$$f_x = 2x + y\cos(xy), \qquad f_y = x\cos(xy).$$

Continuous on $\mathbb{R}^2$ because polynomials and cos function is continuous everywhere and sum and multiplication of continuous functions are continuous again.

Problem 5

$f(x,y,z) = \dfrac{xy}{1+z}$. Compute f_x, f_y, f_z and state where they are continuous.

Solution.

$$f_x = \frac{y}{1+z}, \qquad f_y = \frac{x}{1+z}, \qquad f_z = \frac{-xy}{(1+z)^2}.$$

Continuous for $z \neq -1$.

Problem 6

$f(x,y) = \sqrt{x^2+y^2}$. Compute f_x, f_y and state where they are continuous.

Solution. For $(x,y) \neq (0,0)$,

$$f_x = \frac{x}{\sqrt{x^2+y^2}}, \qquad f_y = \frac{y}{\sqrt{x^2+y^2}}.$$

At the origin: $\lim\limits_{h\to 0} \dfrac{f(h,0)-f(0,0)}{h} = \lim\limits_{h\to 0} \dfrac{|h|}{h} = \pm 1$, which means partial derivatives do not exist at $(0,0)$. Continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Problem 7

$f(x,y) = \dfrac{x^4+y^4}{x^2+y^2}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$. Compute $f_x(0,0)$ and decide whether f_x is continuous at the origin.

Solution.

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0)}{h} = \lim_{h\to 0} \frac{h^2}{h} = 0.$$

$$f_x = \frac{4x^3(x^2+y^2) - (x^4+y^4)(2x)}{(x^2+y^2)^2} = \frac{2x\,(x^4+2x^2y^2-y^4)}{(x^2+y^2)^2}.$$

Polar $x = r\cos\theta,\ y = r\sin\theta$:

$$f_x = 2r\cos\theta\,(\cos^4\theta + 2\cos^2\theta\sin^2\theta - \sin^4\theta).$$

Since $-1 \le \cos\theta, \sin\theta \le 1$, each power is ≤ 1, so $|\cos^4\theta + 2\cos^2\theta\sin^2\theta - \sin^4\theta| \le 1+2+1 = 4$ and $|\cos\theta| \le 1$:

$$0 \le |f_x| \le 2r \cdot 1 \cdot 4 = 8r \quad \text{then} \lim_{(x,y)\to(0,0)} f_x = 0 = f_x(0,0).$$

f_x continuous on $\mathbb{R}^2$.

Problem 8

$f(x,y) = \dfrac{x^2-y^2}{\sqrt[3]{x^2+y^2}}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$. Compute $f_x(0,0)$ and decide whether f_x is continuous at the origin.

Solution.

$$f(h,0) = \frac{h^2}{(h^2)^{1/3}} = h^{4/3}, \qquad f_x(0,0) = \lim_{h\to 0} \frac{h^{4/3}}{h} = \lim_{h\to 0} h^{1/3} = 0.$$

With $f = (x^2-y^2)(x^2+y^2)^{-1/3}$,

$$f_x = 2x(x^2+y^2)^{-1/3} - \tfrac{2x}{3}(x^2-y^2)(x^2+y^2)^{-4/3} = \frac{4x\,(x^2+2y^2)}{3\,(x^2+y^2)^{4/3}}.$$

Polar: $f_x = \frac{4}{3}\cos\theta\,(\cos^2\theta + 2\sin^2\theta)\,r^{1/3}$. Since $-1 \le \cos\theta, \sin\theta \le 1$: $|\cos\theta| \le 1$ and $|\cos^2\theta + 2\sin^2\theta| \le 1 + 2 = 3$, so

$$0 \le |f_x| \le \tfrac{4}{3} \cdot 1 \cdot 3 \cdot r^{1/3} = 4\,r^{1/3} \quad \text{which follows} \lim_{(x,y)\to(0,0)} f_x = 0 = f_x(0,0).$$

f_x continuous on $\mathbb{R}^2$.

Problem 9

$f(x,y) = \dfrac{x^3 - y^3}{x^2 + y^2}$ for $(x,y) \neq (0,0)$, $f(0,0) = 0$. Compute $f_x(0,0)$ and decide whether f_x is continuous at the origin.

Solution.

$$f(h,0) = \frac{h^3}{h^2} = h, \qquad f_x(0,0) = \lim_{h\to 0} \frac{h}{h} = 1.$$

$$f_x = \frac{3x^2(x^2+y^2) - (x^3 - y^3)(2x)}{(x^2+y^2)^2} = \frac{x^4 + 3x^2y^2 + 2xy^3}{(x^2+y^2)^2}.$$

$$x\text{-axis: } f_x(x,0) = \frac{x^4}{x^4} = 1, \qquad y\text{-axis: } f_x(0,y) = \frac{0}{y^4} = 0, \qquad 1 \neq 0.$$

Limit does not exist, which means f_x not continuous at the origin.

Problem 10

Can any function have $f_x = y^2$ and $f_y = x$?

Solution.

$$f_{xy} = \frac{\partial}{\partial y}(y^2) = 2y, \qquad f_{yx} = \frac{\partial}{\partial x}(x) = 1, \qquad 2y \neq 1.$$

$f_{xy} \neq f_{yx}$ (both continuous) which is a contradiction $\Rightarrow$ no such f exist.

Problem 11

Find f_{xy} for $f(x,y) = x^3y^2 + e^y \sin(x) + x\cos x$.

Solution. $f_{xy} = f_{yx}$; take $\partial/\partial y$ first:

$$f_y = 2x^3y + e^y \sin x, \qquad f_{xy} = f_{yx} = \frac{\partial}{\partial x} f_y = 6x^2y + e^y \cos x.$$

Note. Differentiating in y giving a short f_y and first taking partial derivative with respect to x first would instead require the product rule on $x\cos x$.

Problem 12

True or false: if $f_x = 0$ at every point of its domain, then f depends only on y.

Solution. On $D = \{(x,y) : x \neq 0\}$:

$$f(x,y) = \begin{cases} 0, & x < 0 \\ 1, & x > 0 \end{cases} \qquad f_x = 0 \text{ on } D, \quad f \neq h(y).$$

which is a counter example, f clearly depends on x but the $f_x = 0$ for every value in the domain f.